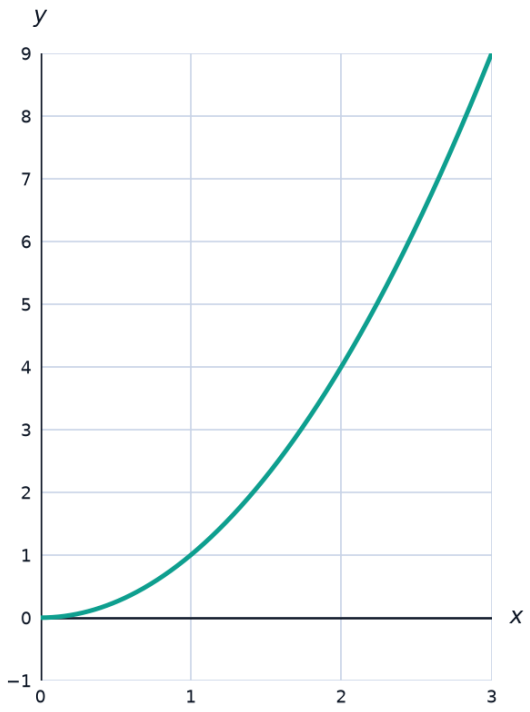
Advanced Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • **Target: Inverse functions with domain restrictions**

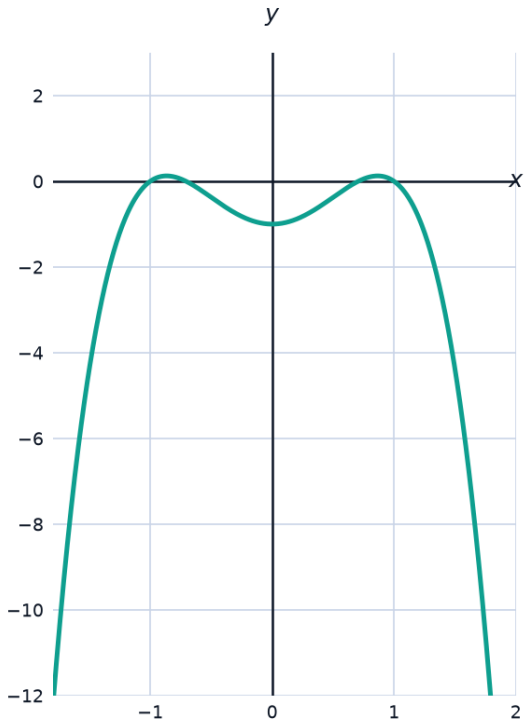
Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.



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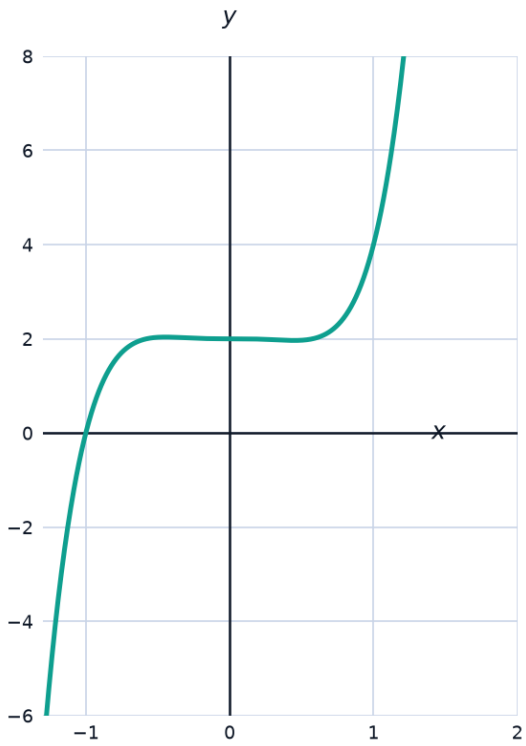
2 [Foundational] • Target: End behavior of polynomial functions

Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \to \infty$ and $x \to -\infty$.



3 [Foundational] • Target: End behavior of polynomial functions

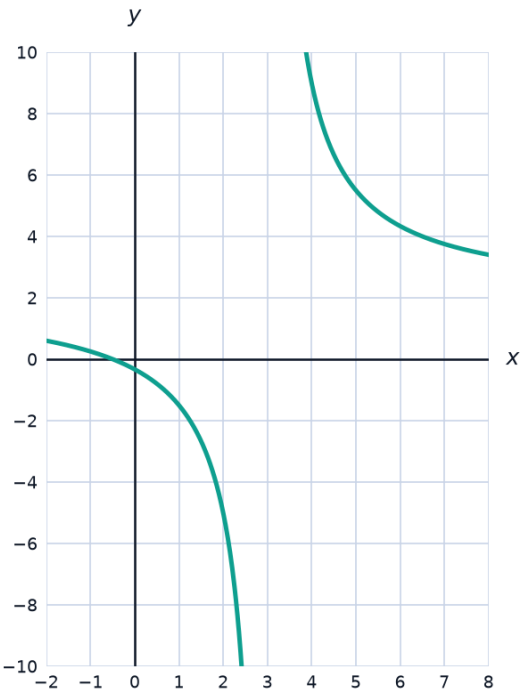
Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.



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4 [Foundational] • Target: Rational functions: asymptotes and holes

Find the vertical and horizontal asymptotes of $f(x) = \frac{2x+1}{x-3}$.



5 [Foundational] • Target: Even, odd, and periodic function classification

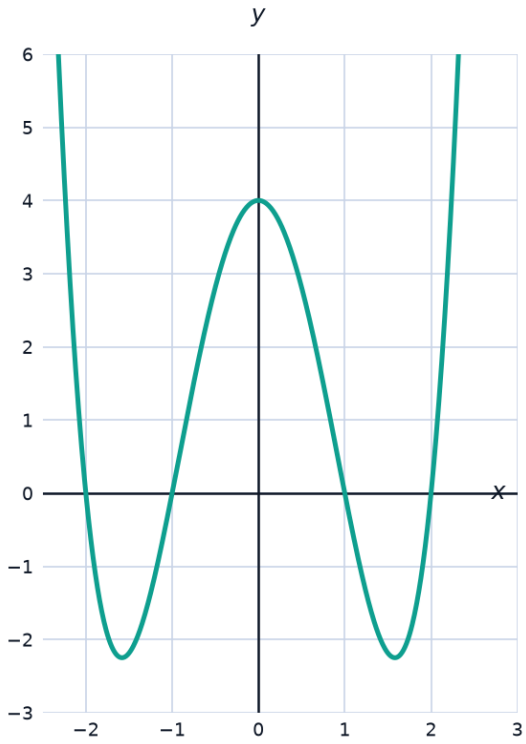
Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.



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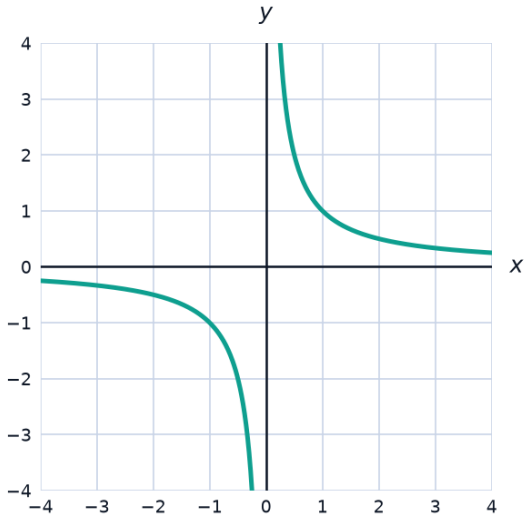
6 [Foundational] • Target: Finding zeros of higher-degree functions graphically and algebraically

A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.



7 [Foundational] • Target: Symmetry and transformations of rational functions

Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.

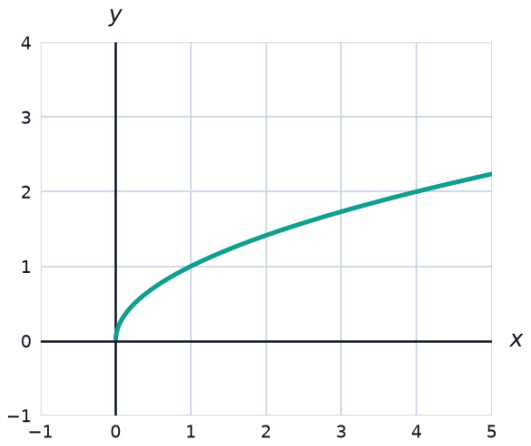


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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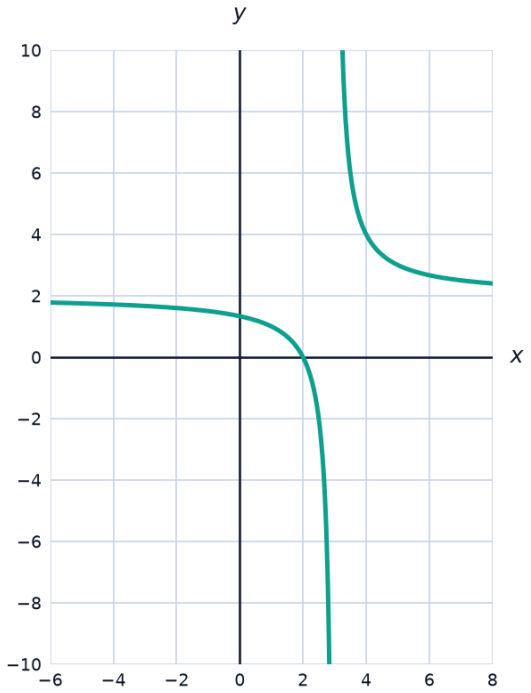
8 **[Intermediate]** • **Target: Finding a function's formula from a description of transformations**

The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.



9 **[Intermediate]** • **Target: A multi-part rational function analysis problem**

This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.



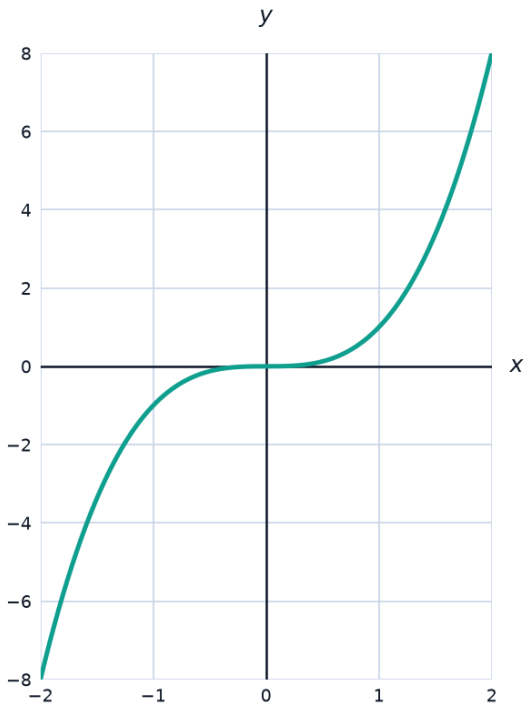
10 **[Intermediate]** • **Target: Finding a quadratic function from a table of values**

A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a, b, and c.

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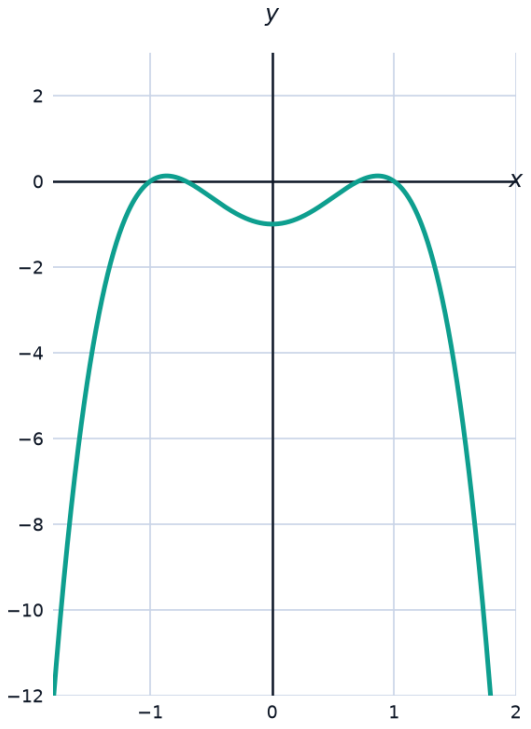
11 **[Intermediate]** • **Target: Determining if a function has an inverse using the horizontal line test**

Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.



12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

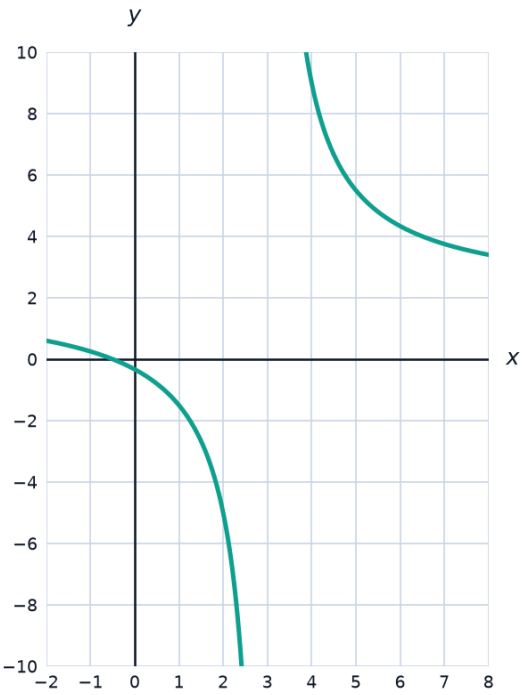
Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \to \infty$ and $x \to -\infty$.



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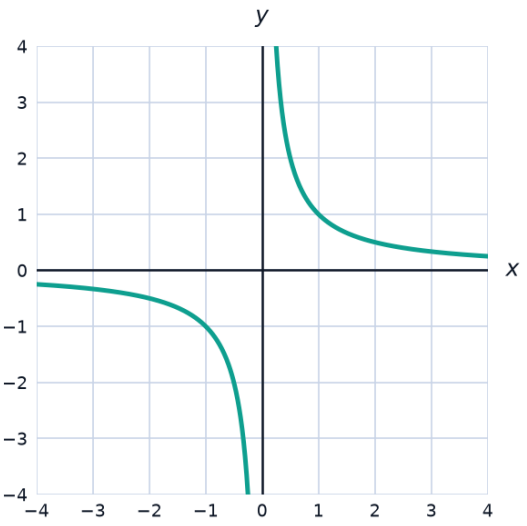
13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the vertical and horizontal asymptotes of $f(x) = \frac{2x+1}{x-3}$.



14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.

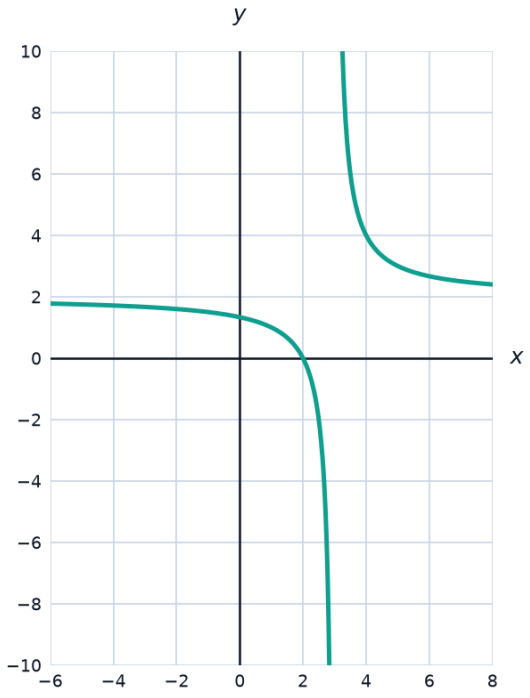


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Trigonometry

This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.



16 [Challenging] • Target: Composite functions with restrictions • Cross-Lesson: 5.01 Pre-Calculus Concepts

If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.

17 [Challenging] • Target: Composite functions with restrictions • Cross-Lesson: 6.01 Advanced Coordinate Geometry

If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.

18 [Challenging] • Target: Rational functions: asymptotes and holes • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2 - 4}{x - 2}$.

19 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 8.01 Statistics, Counting & Probability

Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.

20 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 1.01 Advanced Number Theory

Solve $\ln(x) + \ln(x - 3) = \ln(10)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Composite functions with restrictions

Describe the end behavior of $f(x) = 4x^4 - x^3$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, \infty$

b. $\infty, -\infty$

c. $-\infty, -\infty$

d. ∞, ∞

22 [Foundational] • Target: Inverse functions with domain restrictions

Describe the end behavior of $f(x) = -x^3 + 8x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $\infty, -\infty$

b. $-\infty, \infty$

c. ∞, ∞

d. $-\infty, -\infty$

23 [Foundational] • Target: End behavior of polynomial functions

Describe the end behavior of $f(x) = -4x^4 - 2x^2$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, \infty$

b. $\infty, -\infty$

c. ∞, ∞

d. $-\infty, -\infty$

24 [Foundational] • Target: Rational functions: asymptotes and holes

Describe the end behavior of $f(x) = x^2 - 2x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. ∞, ∞

b. $-\infty, -\infty$

c. $-\infty, \infty$

d. $\infty, -\infty$

25 [Foundational] • Target: Logarithmic functions and their properties

Describe the end behavior of $f(x) = -4x^2 + 4x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. ∞, ∞

b. $-\infty, -\infty$

c. $\infty, -\infty$

d. $-\infty, \infty$

26 [Foundational] • Target: Even, odd, and periodic function classification

Describe the end behavior of $f(x) = x^5 - x^3$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $\infty, -\infty$

b. $-\infty, \infty$

c. $-\infty, -\infty$

d. ∞, ∞

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27 [Foundational] • Target: Piecewise-defined functions and continuity

Find the vertical asymptote of $f(x) = \frac{x-15}{x+14}$.

a. -13

b. -15

c. 14

d. -14

28 [Foundational] • Target: A multi-part function transformation and composition problem

Find the vertical asymptote of $f(x) = \frac{x-7}{2x-12}$.

a. 7

b. -6

c. 6

d. 5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding zeros of higher-degree functions graphically and algebraically

Find the vertical asymptote of $f(x) = \frac{-8x-1}{7x+7}$.

a. 1

b. -1

c. 0

d. -2

30 [Intermediate] • Target: Function operations: sum, difference, product, quotient

Find the vertical asymptote of $f(x) = \frac{9x-10}{x-7}$.

a. -7

b. 7

c. 6

d. 8

31 [Intermediate] • Target: Symmetry and transformations of rational functions

Find the vertical asymptote of $f(x) = \frac{5x+4}{-5x-15}$.

a. -4

b. 3

c. -3

d. -2

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32 [Intermediate] • Target: Finding a function's formula from a description of transformations

Find the vertical asymptote of $f(x) = \frac{x-12}{7x-14}$.

a. -2

b. 2

c. 3

d. 1

33 [Intermediate] • Target: A multi-part rational function analysis problem

Solve $\ln(x) + \ln(x-5) = \ln(55)$.

a. 11

b. 5

c. 10

d. 12

34 [Intermediate] • Target: Finding a quadratic function from a table of values

Solve $\ln(x) + \ln(x-8) = \ln(32)$.

a. 3

b. 8

c. 4

d. 5

35 [Intermediate] • Target: Determining if a function has an inverse using the horizontal line test

The quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.

a. $-2, -1, 1, 2$

b. $-1, -2, 1, 2$

c. $1, 2$

d. $-2, 2$

36 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quartic function $f(x) = x^4 - 17x^2 + 16$ can be treated as a quadratic in x^2. Find all real zeros.

a. $-1, -4, 1, 4$

b. $1, 4$

c. $-4, -1, 1, 4$

d. $-4, 4$

37 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $f(x) = -3x + 5$ and $g(x) = x^2 + 5$, find $(f \circ g)(3)$.

a. -39

b. -37

c. -35

d. 14

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $f(x) = 2x - 1$ and $g(x) = -2x^2$, find $(f \circ g)(-1)$.

a. -7

b. -2

c. -3

d. -5

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Describe the end behavior of $f(x) = -x^3 + 8x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $\infty, -\infty$

b. $-\infty, \infty$

c. ∞, ∞

d. $-\infty, -\infty$

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

Describe the end behavior of $f(x) = 4x^5 + 5x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, -\infty$

b. $\infty, -\infty$

c. ∞, ∞

d. $-\infty, \infty$

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Describe the end behavior of $f(x) = -3x^2 - 9x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, \infty$

b. $\infty, -\infty$

c. $-\infty, -\infty$

d. ∞, ∞

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Trigonometry

Find the vertical asymptote of $f(x) = \frac{2x+11}{-x-8}$.

a. -7

b. 8

c. -9

d. -8

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the vertical asymptote of $f(x) = \frac{5x + 4}{-5x - 15}$.

a. -4

b. 3

c. -3

d. -2

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the vertical asymptote of $f(x) = \frac{7x + 14}{3x - 9}$.

a. 2

b. 4

c. -3

d. 3

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\log_b 3 = x$ and $\log_b 5 = y$, what is $\log_b (45)$ in terms of x and y?

a. $2x + y$

b. $x^2 + y$

c. $2xy$

d. $x + 2y$

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